

Pouyan Navard

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About Me

Generative AI researcher building efficient, controllable diffusion models for video, image, and 3D generation—with industry experience in diffusion-based synthetic data generation for physical AI and robotic perception.

Skills

Generative AI & Diffusion Models: Video Diffusion Models, Diffusion Transformers (DiT), Stable Diffusion, ControlNet, Synthetic Data Generation, LoRA/QLoRA/PEFT, DPO for Generative Models, Image/Video/3D Generation, 3D Morphable Models (FLAME)

Vision & Multimodal AI: Vision Transformers, Vision-Language Models (LLaVA), Masked Multimodal Modeling, 3D Perception, Point Cloud Processing, Physical AI

Frameworks & Infrastructure: PyTorch, HuggingFace, PyTorch Lightning, CUDA, DDP/FSDP/DeepSpeed, NCCL, Multi-Node GPU Training, Hydra, OpenCV, Open3D, SLURM, AWS, Weights & Biases, Git

Languages: Python, C++, Bash

Industry Experience

Computer Vision Engineer

Path Robotics Inc. [🔗](#)

Columbus, Ohio

Nov 2024 – May 2026

- Designed a **mesh diffusion pipeline** for synthetic training data generation in a robotic welding system, with controllable lighting, texture, and geometry—producing photorealistic 3D assets at ~2 min/asset on H100 and eliminating the need for real-world data collection (**86.7%** reduction), replacing a \$15/hr manual annotation pipeline with near-zero-cost compute.
- Architected a ~800M-parameter **multimodal foundation model** trained via cross-modal masked modeling on paired point cloud and image data across multi-GPU clusters, consolidating 5 per-customer models into a single generalizable model with **96%** classification accuracy.
- Built a Hydra-based MLOps framework with SLURM-managed distributed training and Weights & Biases tracking, adopted as the standard template across research teams.

Selected Publications

Loki: Representation over Architecture for Diffusion-Based Portrait Animation [🔗](#)

Pouyan Navard, Sernam Lim

Conference on Neural Information Processing Systems (NeurIPS), 2026 (submitted)

- Developed a **video diffusion model** for portrait animation that conditions on FLAME parametric face representations instead of RGB, achieving controllable cross-identity reenactment without cross-ID training data while requiring **43% fewer** parameters and **1,496× less** training video than the leading baseline. Trained on 8×H200 GPUs.
- Introduced two evaluation metrics (Head Pose Follow, Expression Follow) grounded in the parametric face model, exposing systematic gaps in the field's inherited metric suite.

KnobGen: Controlling the Sophistication of Artwork in Sketch-Based Diffusion Models [🔗](#)

Pouyan Navard, Amin Monsefi, Mengxi Zhou, Wei-Lun Chao, Alper Yilmaz, Rajiv Ramnath

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Workshop, 2025

- Proposed a dual-pathway diffusion framework for **multi-condition controllable** image generation from sketches, with coarse-grained and fine-grained spatial controllers linked by an adjustable knob mechanism, enabling high-quality synthesis from both novice and expert inputs.

SeamCam: Quantifying Seamless Camouflage via Multi-Cue Visual Detectability

Amin Monsefi et al.

European Conference on Computer Vision (ECCV), 2026 (under review)

- Applied **Direct Preference Optimization (DPO) fine-tuning** to diffusion-based inpainting models for

camouflage image generation; introduced a camouflage evaluation metric achieving 78.82% agreement with human judgments and a 1.5k-image benchmark.

Large Language-and-Vision Assistant for Lunar Exploration (LLaVA-LE) [↗](#)

Gokce Inal, **Pouyan Navard**, Alper Yilmaz

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Workshop, 2026

- Fine-tuned LLaVA for lunar surface characterization via two-stage instruction tuning, achieving a **3.3**× performance gain; curated LUCID, a 96k-image multimodal dataset with 81k QA pairs from NASA archives.

SegFormer3D: An Efficient Transformer for 3D Medical Image Segmentation [↗](#)

Shehan Perera, **Pouyan Navard**, Alper Yilmaz

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Workshop, 2024

- Designed a lightweight hierarchical transformer achieving competitive accuracy with **33**× **fewer** parameters and **13**× **lower** compute than state-of-the-art 3D segmentation models.

Full publication list: [Google Scholar](#) [↗](#)

Awards and Honors

Robert E. Altenhofen Memorial Scholarship Award [↗](#)

ISPRS, 2022

- National-level scholarship recognizing excellence in photogrammetry and remote sensing research.

Education

The Ohio State University

Feb 2021 – Aug 2025

PhD in Electrical and Computer Engineering

- GPA: 3.9/4.0
- Dissertation: *Distribution-Aware Learning for Sparsely Labeled and Imbalanced Spatiotemporal Data*
- Research focus on 3D perception, diffusion-based generative modeling, and multimodal learning
- Supervisor: Prof. Alper Yilmaz.

Isfahan University

Sep 2015 – Sep 2017

MSc in Electrical and Computer Engineering

- GPA: 3.9/4.0
- Thesis: Geometric Structure from Motion under Large View Point Change using Epipolar Geometry

Isfahan University

Sep 2011 – Sep 2015

BSc in Electrical and Computer Engineering

- GPA: 3.6/4.0
- Thesis: 3D View Photogrammetric Reconstruction